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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I -III Semester(2025-26)
Faculty Name	Ms. Alluri Swathi
Student Name	Sivarapu Rishil
Roll Number	2520090174
Branch	CSIT
Course Outcome	1

Case Study Topic: UrbanAI – Intelligent Smart City Infrastructure Optimization Platform using AVL tree

Work Done Summary:
Implemented AVL Tree in Java to manage smart city infrastructure records using Asset IDs. Added operations for inserting, searching, deleting, and displaying infrastructure assets such as traffic signals, metro stations, road sensors, and emergency units. Implemented AVL balancing techniques to maintain efficient searching and avoid skewed tree structures. Used inorder traversal to display infrastructure records in sorted order and generated smart city analytics such as total active assets.
Compared AVL Tree performance with ArrayList searching and observed better efficiency for large-scale smart city data management.

Implementation Details :
1- Created Node class with fields: assetId, infrastructureName, location, maintenanceStatus, left, right, and height.
2- Insert() adds infrastructure records into the AVL Tree.
3- AVL rotations maintain balanced tree structure for efficient operations.
4- Search() finds infrastructure assets using Asset IDs.
5- Delete() removes inactive or duplicate infrastructure records.
6- InorderTraversal() displays infrastructure records in sorted order.
7- Analytics methods calculate total active infrastructure assets.
8- The main method demonstrated insertion, searching, deletion, traversal, and smart city analytics operations.

Result : Screenshots/Output:

```
====== URBANAI INFRASTRUCTURE RECORDS ======


--------------------------------
Asset ID            : 95
Infrastructure Name : Road Sensor
Location            : Highway Sector
Status              : Under Maintenance
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Asset ID            : 120
Infrastructure Name : Traffic Signal
Location            : Central Junction
Status              : Active
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Asset ID            : 180
Infrastructure Name : Smart Street Light
Location            : Downtown
Status              : Operational
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Asset ID            : 210
Infrastructure Name : Metro Station
Location            : North Zone
Status              : Operational
--------------------------------
Asset ID            : 300
Infrastructure Name : Emergency Control Unit
Location            : City Hospital
Status              : Active

====== SEARCH RESULT ======

Infrastructure Asset Found!
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Asset ID            : 210
Infrastructure Name : Metro Station
Location            : North Zone
Status              : Operational

Deleting Asset ID 95...
```

```
====== UPDATED RECORDS ======

--------------------------------
Asset ID            : 120
Infrastructure Name : Traffic Signal
Location            : Central Junction
Status              : Active
--------------------------------
Asset ID            : 180
Infrastructure Name : Smart Street Light
Location            : Downtown
Status              : Operational
--------------------------------
Asset ID            : 210
Infrastructure Name : Metro Station
Location            : North Zone
Status              : Operational
--------------------------------
Asset ID            : 300
Infrastructure Name : Emergency Control Unit
Location            : City Hospital
Status              : Active

====== SMART CITY ANALYTICS ======

Total Active Infrastructure Assets : 4
```

Time Complexity:

- AVL Tree Insert Operation: O(log n)
- AVL Tree Delete Operation: O(log n)
- AVL Tree Search Operation: O(log n)
- Inorder Traversal for Infrastructure Reports: O(n)
- Infrastructure Analytics Calculation: O(n)
- AVL Rotations (Left/Right Rotations): O(1)
- Compared to ArrayList searching O(n), AVL Trees provide faster and balanced searching, insertion, and deletion operations for large-scale smart city infrastructure management systems.

GitHub Link : https://github.com/rishill10/DSA2_CO1

Student Signature	Faculty Signature